

OPTIMAL ADAPTIVE OUTPUT REGULATION OF DISCRETE-TIME NONLINEAR STOCHASTIC SYSTEMS*

ZHAOBO LIU[†] AND CHANYING LI[‡]

Abstract. This paper addresses the output regulation problem associated with a basic class of discrete-time nonlinear stochastic systems with unknown parameters. We allow the controlled plants to exhibit highly nonlinear growth, as well as nonlinearly parameterized structures in the exosystems. Under certain mild conditions, it is shown that with stable exosystems, the closed-loop regulated outputs are asymptotically optimal in the average sense almost surely. Technologically, we derive a near-optimal convergence rate for the nonlinear least squares (NLS) problem when identifying exosystems. In particular, upon degeneration of exosystems to null, we actually extend the existing stabilizability theorems to a more general setting.

Key words. discrete-time, nonlinear least squares, nonlinear systems, output regulation, stochastic adaptive control

MSC codes. 93D15, 93E12, 93E24

DOI. 10.1137/24M1679197

1. Introduction. Output regulation is a fundamental problem in control theory, aiming to control a designated plant to track external references or counteract disturbances, originating from an exosystem. Initially considered for linear deterministic systems, output regulation leverages the widely recognized internal model principle [6, 11]. Over time, the scope of output regulation expanded to include a wide range of nonlinear systems [16, 19, 29] and infinite-dimensional systems [12]. A general framework to address nonlinear output regulation is to transform the original system into an augmented system, comprising the plant and a constructed internal model [18]. This transformation simplifies the problem to the stabilization of the augmented system.

There inherently exist uncertainties in modeling real-world systems, which makes it impractical to precisely solve the nonlinear regulator equations and construct satisfactory internal models. This limitation has spurred interest in adaptive construction methods for internal models. For example, linearly parameterized internal models have been constructed in [33, 34], while [3, 4, 10] have considered the adaptive approach for nonlinear internal models as a system identification problem. But most existing studies primarily concentrate on the deterministic case, despite the prevalence of stochastic modeling in engineering control. Considering that computer simulation

*Received by the editors August 5, 2024; accepted for publication (in revised form) March 31, 2025; published electronically July 7, 2025.

<https://doi.org/10.1137/24M1679197>

Funding: This work was supported in part by the National Natural Science Foundation of China under grants 12401585, 11925109, and U21B6001, by the National Science Foundation of the Guangdong Province of China under grant 2024A1515011542, by the Guangdong Provincial Ordinary Universities Youth Innovative Talent Project (Natural Science) under grant 2023KQNCX064, and by the Scientific Foundation for Youth Scholars of Shenzhen University under grant 868-000001032170.

[†]Institute for Advanced Study, Shenzhen University, Shenzhen, 518060 People's Republic of China (liuzhaobo@szu.edu.cn).

[‡]Corresponding author. The Key Laboratory of Systems and Control, Academy of Mathematics and Systems Science, Chinese Academy of Sciences, Beijing, 100190 People's Republic of China, and the School of Mathematical Sciences, University of Chinese Academy of Sciences, Beijing, 100049, People's Republic of China (cyli@amss.ac.cn).

Enhancing Multivariate Time Series Forecasting: A Novel Approach with Mallows Model Averaging and Graph Neural Networks*

ZHANG Haili · WANG Jiawei · LIU Zhaobo · DONG Hailing

DOI: 10.1007/s11424-024-4044-9

Received: 31 January 2024 / Revised: 25 April 2024

©The Editorial Office of JSSC & Springer-Verlag GmbH Germany 2025

Abstract Multivariate time series forecasting holds substantial practical significance, facilitates precise predictions, and informs decision-making. The complexity of nonlinear relationships and the presence of higher-order features in multivariate time series data have sparked a burgeoning interest in leveraging deep learning approaches for such forecasting tasks. Existing methods often use pre-scaled neural networks, whose reliability and generalization can pose a challenge. In this study, the authors propose an instance-wise graph-based Mallows model averaging (IGMMA) framework for multivariate time series prediction. The framework incorporates a model averaging module into the network, where extracted features are utilized as inputs for candidate linear models. These linear models are combined with weights to create a new linear layer, forming a novel graph neural network model. Moreover, the network loss function is modified based on the Mallows criterion, where penalties are imposed on the parameters and the weights separately. The authors use the proposed method to predict multicommodity futures prices, and the empirical results show that IGMMA has superior predictive accuracy even when small neural networks are used. This indicates that the model averaging module significantly reduces the parameters required for deep learning training, which enables the training of multiple small models as an alternative to training a large model.

Keywords Graph neural networks, model averaging, multivariate time series.

ZHANG Haili

School of Undergraduate Education, Shenzhen Polytechnic University, Shenzhen 518055, China.

Email: zhanghl@szpu.edu.cn.

WANG Jiawei · LIU Zhaobo (Corresponding author)

Institute for Advanced Study, Shenzhen University, Shenzhen 518060, China.

Email: iwnlgre@outlook.com; liuzhaobo@szu.edu.cn.

DONG Hailing

Science of Mathematics College, Shenzhen University, Shenzhen 518060, China. Email: hailing@szu.edu.cn.

*This research was supported by the Research Foundation of Shenzhen Polytechnic University under Grant No. 6023312034K, the Post-doctoral Later-Stage Foundation Project of Shenzhen Polytechnic University under Grant No. 6023271021K, the National Science Foundation of Guangdong Province of China under Grant No. 2024A1515011542, the Guangdong Provincial Ordinary Universities Youth Innovative Talent Project (Natural Science) under Grant No. 2023KQNCX064, the National Natural Science Foundation of China under Grant No. 12371448, and the National Key R&D Program of China under Grant No. 2023YFE0126800.

◇ This paper was recommended for publication by Editor LIAO Jun.

Optimal Linear Crossover for Mitigating Negative Transfer in Evolutionary Multitasking

Zhaobo Liu¹, Jianhua Yuan², Haili Zhang, Tao Zeng, and Zexuan Zhu³, *Senior Member, IEEE*

Abstract—Evolutionary multitasking (EMT) algorithms use information exchange among individuals in a population to solve multiple optimization problems simultaneously. Negative transfer is a critical factor that affects the performance of EMT algorithms. In this study, we propose an innovative approach to mitigate negative transfer in EMT algorithms. The proposed approach is grounded in rigorous theoretical analysis, which provides valuable theoretical insights into the design of an optimal linear crossover operator for mitigating negative transfer. By identifying interpretable conditions, we establish a solid theoretical foundation to prevent negative transfer in diverse scenarios. Building upon these findings, we theoretically derive a closed-form expression for the optimal crossover (OC) operator and propose practical design methods based on approximations. Furthermore, we integrate the proposed OC operator into a fundamental EMT algorithm framework. The resultant algorithm is comparable or superior to other state-of-the-art methods. Empirical validation through comprehensive experiments confirms the effectiveness of our theoretical findings.

Index Terms—Evolutionary algorithms (EAs), multitask optimization (MTO), optimal crossover (OC).

I. INTRODUCTION

EVOLUTIONARY algorithms (EAs) are metaheuristic optimization algorithms inspired by the principles of

biological evolution [1]. EAs simulate the biological evolution process mainly via crossover, mutation, selection, and evaluation. EAs exhibit robust search capabilities and scalability, making them applicable in various research domains, including artificial intelligence, machine learning, environmental management, and complex optimization problems [2], [3], [4], [5], [6], [7]. Within the realm of optimization problems, EAs have spawned numerous innovative and practical algorithms customized for various optimization task requirements. Notably, multitask optimization (MTO), a recent development, have garnered significant attention and research interest in the evolutionary computation community. The primary goal of MTO is to simultaneously optimize multiple problems, fostering mutual influence and knowledge transfer among different tasks to achieve superior results compared with individual task optimization. To tackle this emerging challenge, evolutionary multitasking (EMT) employs EAs to simultaneously solve MTO tasks. By exploiting the advantages of knowledge transfer, EMT has been demonstrated to outperform single-task EAs on various optimization problems and real-world applications.

The multifactorial EA (MFEA) [8] and its variants [9], [10] represent the most widely used EMT algorithms. Fundamentally, the key to design efficient MFEAs is to enhance knowledge transfer efficiency or alleviate the threat of negative transfer. Various methods have been proposed for selecting the most suitable individuals for knowledge transfer. For instance, decision trees have been used in [11] to identify promising individuals that enhance transfer efficiency, while the work [12] introduced the concept of “skill membership” to select individuals best suited for each task, using the convergence rate as a guideline to prevent population from being trapped in local optima. Unlike the process of selecting appropriate individuals, Ma et al. [13] utilized the Kullback-Leibler divergence to calculate a finer-grained genetic similarity. Surrogate models have also been employed to create helper problems with smoother and simpler functions, facilitating efficient knowledge transfer [14], [15]. In MFEA, a parameter known as the random mating probability (rmp) is also usually used to regulate the frequency of knowledge transfer leading to better transfer efficiency. Employing a fixed rmp value without considering the differences between tasks can easily lead to negative transfer. Therefore, adaptive rmp setting approaches have been explored in [16], [17], [18]. Similar to the idea of using adaptive rmp, Xu et al. [19] calculated the probability of knowledge transfer based on historical knowledge, with the aim of controlling the

Received 24 March 2024; revised 21 July 2024 and 22 September 2024; accepted 25 October 2024. Date of publication 4 November 2024; date of current version 2 February 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 12401585, Grant 62471310, and Grant 12401393; in part by the Natural Science Foundation of Guangdong Province of China under Grant 2024A1515011542; in part by the Foundation for Distinguished Young Talents in Higher Education of Guangdong, China, under Grant 2023KQNCX064, and in part by the Open Funding of Key Laboratory of Dependable Service Computing in Cyber Physical Society, Ministry of Education of China under Grant CPSDSC202004. This article was approved by Associate Editor A. Gupta. (Corresponding author: Zexuan Zhu.)

Zhaobo Liu is with the Institute for Advanced Study, Shenzhen University, Shenzhen 518060, China (e-mail: liuzhaobo@szu.edu.cn).

Jianhua Yuan is with the College of Computer Science and Software Engineering, Shenzhen University, Shenzhen 518060, China (e-mail: yuanjianhua2022@email.szu.edu.cn).

Haili Zhang is with the School of Undergraduate Education, Shenzhen Polytechnic University, Shenzhen 518055, China (e-mail: zhanghl@szpu.edu.cn).

Tao Zeng is with the National Engineering Laboratory for Big Data System Computing Technology, Shenzhen University, Shenzhen 518060, China (e-mail: szuztao@163.com).

Zexuan Zhu is with the National Engineering Laboratory for Big Data System Computing Technology, Shenzhen University, Shenzhen 518060, China, and also with the Key Laboratory of Dependable Service Computing in Cyber Physical Society, Ministry of Education of China, Chongqing University, Chongqing 400044, China (e-mail: zhuzx@szu.edu.cn).

This article has supplementary downloadable material available at <https://doi.org/10.1109/TEVC.2024.3490174>, provided by the authors.

Digital Object Identifier 10.1109/TEVC.2024.3490174

Multifactorial Evolutionary Algorithm Based on Diffusion Gradient Descent

Zhaobo Liu¹, Guo Li, Haili Zhang, Zhengping Liang², and Zexuan Zhu³, *Senior Member, IEEE*

Abstract—The multifactorial evolutionary algorithm (MFEA) is one of the most widely used evolutionary multitasking (EMT) algorithms. The MFEA implements knowledge transfer among optimization tasks via crossover and mutation operators and it obtains high-quality solutions more efficiently than single-task evolutionary algorithms. Despite the effectiveness of MFEA in solving difficult optimization problems, there is no evidence of population convergence or theoretical explanations of how knowledge transfer increases algorithm performance. To fill this gap, we propose a new MFEA based on diffusion gradient descent (DGD), namely, MFEA-DGD in this article. We prove the convergence of DGD for multiple similar tasks and demonstrate that the local convexity of some tasks can help other tasks escape from local optima via knowledge transfer. Based on this theoretical foundation, we design complementary crossover and mutation operators for the proposed MFEA-DGD. As a result, the evolution population is endowed with a dynamic equation that is similar to DGD, that is, convergence is guaranteed, and the benefit from knowledge transfer is explainable. In addition, a hyper-rectangular search strategy is introduced to allow MFEA-DGD to explore more underdeveloped areas in the unified express space of all tasks and the subspace of each task. The proposed MFEA-DGD is verified experimentally on various multitask optimization problems, and the results demonstrate that MFEA-DGD can converge faster to competitive results compared to state-of-the-art EMT algorithms. We also show the possibility of interpreting the experimental results based on the convexity of different tasks.

Index Terms—Convergence analysis, diffusion gradient descent (DGD), evolutionary multitasking (EMT), multifactorial evolutionary algorithm (MFEA).

I. INTRODUCTION

EVOLUTIONARY multitasking (EMT) [1], [2] solves multiple optimization tasks simultaneously using evolutionary algorithms (EAs). Traditional EAs only solve a single optimization task; however, many real-world optimization tasks are related, and valuable knowledge gained from solving one task can help solve another related task [3], [4]. By exploiting the advantages of knowledge transfer, it has been demonstrated that EMT can outperform single-task EAs on various optimization problems [5], [6], [7], [8], [9] and real-world applications [10], [11], [12], [13], [14].

The multifactorial EA (MFEA) [1] has received rapidly increasing attention due to its simplicity and efficiency, and numerous improvements and MFEA variants have been surveyed in [15] and [16]. The main concepts of some representative algorithms [17], [18], [19], [20], [21], [22], [23], [24], [25], [26], [27], [28], [29], [30], [31], [32], [33] are summarized in Table I. In addition, the MFEA has been extended to solve many-task problems that, which involve more than two optimization tasks [34], [35], [36], [37], [38], [39], [40], [41], [42], [43], and multiobjective multitask problems [44], [45], [46], [47], [48], [49], [50], [51], [52], [53].

MFEAs have been applied successfully to various complex optimization problems. However, to the best of our knowledge, strict theoretical analyses of the convergence of MFEAs and the benefits of knowledge transfer are scarce. Bali et al. [20] presented a pioneering attempt to prove the algorithm's convergence and analyze the effects of intertask interactions in MFEA using a probability distribution to model the target population. Here, the convergence of the probability distribution requires updating the probability density at each point in the search space, which strictly implies that the entire space must be searched with an infinite-sized population. However, this underlying assumption may be unrealistic in MFEAs. For other EMT algorithms, Bai et al. [54] also presented a convergence-guaranteed multitask gradient descent (GD) algorithm; however, although the convergence proof was effective for convex problems, it may not hold for nonconvex tasks, which are more prevalent in practical applications. In addition, Han et al. [30] performed a convergence analysis of a particle swarm optimization-based EMT algorithm, which suffers from the same issue encountered by [20].

Manuscript received 11 December 2022; revised 6 March 2023 and 8 April 2023; accepted 20 April 2023. Date of publication 9 May 2023; date of current version 10 July 2024. This work was supported in part by the National Key Research and Development Program of China under Grant 2022YFF1202104; in part by the National Natural Science Foundation of China under Grant 61871272; in part by the Guangdong Basic and Applied Basic Research Foundation under Grant 2021A1515011911; in part by the Shenzhen Science and Technology Program under Grant JCYJ20190808173617147, Grant JCYJ20220531102617039, and Grant 20200811181752003; and in part by the Open Project of BGIShenzhen under Grant BGIRSZ20200002. This article was recommended by Associate Editor A. M. Abu-Mahfouz. (Corresponding author: Zexuan Zhu.)

Zhaobo Liu is with the College of Computer Science and Software Engineering and the Institute for Advanced Study, Shenzhen University, Shenzhen 518060, China (e-mail: liuzhaobo@szu.edu.cn).

Guo Li and Zhengping Liang are with the College of Computer Science and Software Engineering, Shenzhen University, Shenzhen 518060, China (e-mail: szuliguo@szu.edu.cn; liangzp@szu.edu.cn).

Haili Zhang is with the Institute of Applied Mathematics, Shenzhen Polytechnic, Shenzhen 518055, China (e-mail: zhanghl@szpt.edu.cn).

Zexuan Zhu is with the National Engineering Laboratory for Big Data System Computing Technology, Shenzhen University, Shenzhen 518060, China, also with Shenzhen Pengcheng Laboratory, Shenzhen 518055, China, and also with BGI-Shenzhen, Shenzhen 518083, China (e-mail: zhuzx@szu.edu.cn).

Color versions of one or more figures in this article are available at <https://doi.org/10.1109/TCYB.2023.3270904>.

Digital Object Identifier 10.1109/TCYB.2023.3270904



Further results on inverse eigenvalue problem for mass–spring–inerters systems[☆]

Zhaobo Liu^a, Qida Xie^{b,c}, Chanying Li^{b,c,*}

^a Institute for Advanced Study, Shenzhen University, Shenzhen 518060, PR China

^b Key Laboratory of Systems and Control, Academy of Mathematics and Systems Science, Chinese Academy of Sciences, 100190, PR China

^c School of Mathematical Sciences, University of Chinese Academy of Sciences, Beijing 100049, PR China

ARTICLE INFO

Communicated by A. Morassi

Keywords:

Inverse eigenvalue problem
Natural frequency assignment
Mass–spring–inerters system
Multiple eigenvalues

ABSTRACT

This paper is concerned with the inverse eigenvalue problem (IEP) for “fixed–fixed” mass–spring–inerters systems. Unlike the “fixed–free” case, a “fixed–fixed” system has its both ends attached to the ground. This brings some essential differences in the IEP. We find that the construction strategy proposed in Liu et al. (2022) cannot be readily applied here. So we endeavor to develop a new strategy in this paper to solve the problem and accordingly derive a necessary and sufficient condition for the “fixed–fixed” case.

1. Introduction

Natural frequency assignment is a basic problem in classical vibration theory. It is often mathematically associated with the reconstruction of specific structured matrices from designated spectral data [1]. A typical inverse eigenvalue problem (IEP) in this regard concerns a mass–spring system, in which each mass is connected to its neighbor by a spring. Accordingly, the matrices to be sought are the mass and stiffness matrices. This IEP was solved by translating it into an inverse problem for a Jacobi matrix.

A prominent property of a Jacobi matrix is that its eigenvalues are distinct and strictly interlace the eigenvalues of its truncated matrix. Conversely, [2,3] proved that given a set of interlacing real numbers $0 \leq \lambda_1 < \mu_1 < \lambda_2 < \dots < \mu_{n-1} < \lambda_n$, there exists a unique Jacobi matrix such that $\{\lambda_i\}_{i=1}^n$ and $\{\mu_i\}_{i=1}^{n-1}$ are the eigenvalues of the Jacobi matrix and its truncated matrix, respectively. It was based on these results that researchers recovered a mass–spring system in the IEP [4,5]. More precisely, they untangled the mass and stiffness matrices from a Jacobi matrix, which was constructed according to the prescribed eigenvalues. Variants of IEPs for Jacobi operators and modified mass–spring systems are also addressed in the literature [5–8]. Among these studies, it is noteworthy that for physical realization, the prescribed eigenvalues must be distinct due to the inherent property of Jacobi matrices mentioned above.

But the situation changes when involving inerters. The inerter is a new mechanical element first proposed to complete the force–current analogy between mechanical and electrical systems in [9]. Owing to the special merits in practice, inerters are applied to many engineering fields such as building vibration control, landing gears, vibration isolators, train suspensions, vehicle suspensions, and so on [10–15]. Recall that the masses should be appropriately chosen in the IEP of a mass–spring system. But when inerters are introduced in parallel to the springs, [16] found that as long as the given eigenvalues are positive and distinct, the IEP for

[☆] This work is supported by the National Key R&D Program of China (2018YFA0703800), the National Natural Science Foundation of China under Grant 11925109 and CAS Project for Young Scientists in Basic Research under Grant YSBR-008.

* Corresponding author at: Key Laboratory of Systems and Control, Academy of Mathematics and Systems Science, Chinese Academy of Sciences, 100190, PR China.

E-mail address: cyl@amss.ac.cn (C. Li).

Asymptotic behavior of least squares estimators for nonlinear autoregressive models

Zhaobo LIU¹ & Chanying LI^{2,3*}

¹College of Computer Science and Software Engineering, Shenzhen University, Shenzhen 518060, China;

²Key Laboratory of Systems and Control, Academy of Mathematics and Systems Science, Chinese Academy of Sciences, Beijing 100190, China;

³School of Mathematical Sciences, University of Chinese Academy of Sciences, Beijing 100049, China

Received 27 January 2021/Revised 11 May 2021/Accepted 30 June 2021/Published online 17 November 2022

Citation Liu Z B, Li C Y. Asymptotic behavior of least squares estimators for nonlinear autoregressive models. Sci China Inf Sci, 2023, 66(6): 169204, <https://doi.org/10.1007/s11432-021-3309-7>

Dear editor,

Estimations of nonlinear autoregressive (AR) models in the literature typically involve ergodic series. Based on this assumption, the asymptotic theory has been established accordingly (see [1–3]). However, this good property is not always true [4]. For example, we consider

$$y_{t+1} = \theta^\top \phi(y_t, \dots, y_{t-n+1}) + w_{t+1}, \quad t \geq 0, \quad (1)$$

where θ is the $m \times 1$ unknown parameter vector and y_t and w_t are the scalar observations and random noise signals, respectively. Moreover, $\phi: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is a known Lebesgue measurable vector function:

$$\phi(z_1, \dots, z_n) = \text{col}\{\phi^{(1)}(z_1), \dots, \phi^{(n)}(z_n)\}, \quad (2)$$

where $\phi^{(i)} = (f_{i1}, \dots, f_{im_i})^\top: \mathbb{R} \rightarrow \mathbb{R}^{m_i}, i = 1, \dots, n$ are certain known Lebesgue measurable vector functions and $m_i \geq 1$ are n integers satisfying $\sum_{i=1}^n m_i = m$. Without loss of generality, let $y_t = 0$ for $t < 0$. Clearly, most functions ϕ produce non-ergodic sequences $\{y_t\}$. Therefore, in this article, parameter θ in model (1), whose outputs are not necessarily ergodic, is identified.

Least squares (LS) estimators are known as the most efficient algorithms in parameter estimation. Its strong consistency for model (1) depends crucially on matrix $P_{t+1}^{-1} = I_m + \sum_{i=0}^t \phi_i \phi_i^\top$, where $\phi_i = \phi(y_i, \dots, y_{i-n+1})$. Let $\lambda_{\min}(t+1)$ and $\lambda_{\max}(t+1)$ denote the minimal and maximal eigenvalues of P_{t+1}^{-1} , respectively. In the Gaussian case, Ref. [5] showed

$$\left\{ \lim_{t \rightarrow +\infty} \lambda_{\min}(t+1) = +\infty \right\} = \left\{ \lim_{t \rightarrow +\infty} \hat{\theta}_t = \theta \right\}, \quad (3)$$

whereas Ref. [6, Theorem 1] found that when $\{w_t\}$ is an appropriate martingale difference sequence,

$$\|\hat{\theta}_{t+1} - \theta\|^2 = O\left(\frac{\log(\lambda_{\max}(t+1))}{\lambda_{\min}(t+1)}\right), \quad \text{a.s.} \quad (4)$$

* Corresponding author (email: cyli@amss.ac.cn)

Moreover, Ref. [6] pointed out that $\log(\lambda_{\max}(t+1)) = o(\lambda_{\min}(t+1))$ is in some sense the weakest condition for the strong consistency of $\hat{\theta}_t$ in general.

The eigenvalues of P_{t+1}^{-1} depend on outputs $\{y_t\}$, which are produced automatically by the nonlinear random system (1). Thus, checking $\lim_{t \rightarrow +\infty} \lambda_{\min}(t+1) = +\infty$ or $\log(\lambda_{\max}(t+1)) = o(\lambda_{\min}(t+1))$ is nontrivial. However, Ref. [7] successfully verified $\liminf_{t \rightarrow +\infty} t^{-1} \lambda_{\min}(t+1) > 0$, a.s. for the linear AR model,

$$y_{t+1} = \theta_1 y_t + \theta_2 y_{t-1} + \dots + \theta_n y_{t-n+1} + w_{t+1}, \quad t \geq 0, \quad (5)$$

which is a special case of (1). Then, Ref. [7] completely solved the strong consistency of the LS estimator for this basic situation. The proof in [7] attributes to the linear structure of the AR model (5) to a certain extent. Regarding the nonlinear model (1), we aim to identify whether the LS estimator retains a similar asymptotic behavior.

This study establishes the asymptotic properties of the LS estimator for model (1). By assuming mild conditions on ϕ , the minimal eigenvalue of P_{t+1}^{-1} is estimated. We find that the LS estimates converge to the true parameter almost surely on the set where vectors $(y_t, \dots, y_{t-n+1})^\top$ do not diverge to infinity. This means the LS estimator is highly likely to have a strong consistency when applied to model (1) in practice because most real systems are non-divergent. Finally, Appendix A provides the proofs of the main theorems.

Gaussian case. We assume the following conditions.

(A1) Noise $\{w_t\}$ is an i.i.d. random sequence with $w_1 \sim N(0, 1)$, and parameter $\theta \sim N(\theta_0, I_m)$ is independent of $\{w_t\}$.

(A2) Certain open sets $\{E_i\}_{i=1}^n$ belong to $\mathbb{R}$ satisfying the following:

(i) $f_{ij} \in C(\mathbb{R})$ and $f_{ij} \in C^{m_i}(E_i)$, $1 \leq j \leq m_i$, $1 \leq i \leq n$;

(ii) For every unit vector $x \in \mathbb{R}^m$, a point $y \in \prod_{i=1}^n E_i$ exists such that $|\phi^\top(y)x| \neq 0$.

Remark 1. By assumption (A2)(ii), for every unit vector $x \in \mathbb{R}^m$, $\ell(\{y \in \prod_{i=1}^n E_i : |\phi^\top(y)x| > 0\}) > 0$, where ℓ denotes the Lebesgue measure.

Global Stabilizability Theorems on Discrete-Time Nonlinear Uncertain Systems

Zhaobo Liu  and Chanying Li , *Member, IEEE*

Abstract—This article focuses on the stabilizability problem for a basic class of discrete-time nonlinear systems with multiple unknown parameters. We claim that such a system is stabilizable if its nonlinear growth rate is dominated by a polynomial rule. This rule cannot be relaxed in general since it becomes a necessary and sufficient condition when the system has a polynomial form (Li and Lam, 2013). We further prove that the concerned stabilizable system is possible to grow exponentially fast. Meanwhile, optimality and closed-loop identification are also discussed herein.

Index Terms—Discrete time, least squares (LS), nonlinear systems, stabilizability, stochastic adaptive control.

NOMENCLATURE

e.g.	Exempli gratia.
i.o.	Infinitely often.
$\sigma\{y_i, 0 \leq i \leq t\}$	The σ -field generated by $\{y_i, 0 \leq i \leq t\}$.
$I_{\{\cdot\}}$	Indicator function.
$ A $	Determinant of matrix A .
$\ x\ $	Euclidean norm of vector x .
$f(x) = O(g(x))$	There are two constants ζ_1 and $\zeta_2 > 0$ such that for
$(x \rightarrow \infty)$	$ x > \zeta_2, \ f(x)\ \leq \zeta_1 \cdot \ g(x)\ $.

I. INTRODUCTION

ADAPTIVE control of linear systems ([2], [3]) and nonlinear systems growing linearly ([4], [5]) has been a mature topic for decades, both in continuous time and discrete time. But when it comes to systems whose output nonlinearities are faster than linearities, the similarities of adaptive control between continuous- and discrete-time systems disappear. Most continuous-time nonlinear systems can be globally stabilized

by employing nonlinear damping or back-stepping techniques ([6], [7]); however, their discrete-time counterparts are not that favored by fortune. It was found in [8] that even for the following basic discrete-time stochastic system

$$y_{t+1} = \theta y_t^b + u_t + w_{t+1} \quad (1)$$

where θ is a scalar parameter, and y_t , u_t , and w_t are the output, input, and noise signals, respectively, the stabilizability holds if and only if $b < 4$. This fundamental difficulty in discrete-time control was further confirmed in [9], where system (1) is extended to the multiparameter case:

$$y_{t+1} = \theta_1 y_t^{b_1} + \theta_2 y_t^{b_2} + \cdots + \theta_n y_t^{b_n} + u_t + w_{t+1}. \quad (2)$$

An “impossibility theorem” was established therein by providing a polynomial rule, which was proved, a decade later in [1], to be a necessary and sufficient condition of the stabilizability for system (2). This polynomial rule is characterized by $b_1, \dots, b_n$, which shows that the nonlinearity plays a key role in determining the stabilizability. Analogous phenomena arise in adaptive control of various discrete-time nonlinear systems. We refer the readers to [10], [11], [12], [13], [14], [15], and [16].

We are now intended to extend the results of [1] and [17] to the following class of uncertain systems:

$$y_{t+1} = \theta_1 f_1(y_t) + \theta_2 f_2(y_t) + \cdots + \theta_n f_n(y_t) + u_t + w_{t+1}. \quad (3)$$

It is easy to understand that stabilizability depends on closed-loop identifiability. In addition, the more different the functions $f_i(x)$ and $f_j(x)$ are, the easier the parameters θ_i and θ_j can be distinguished (we cannot distinguish θ_i from θ_j if $f_i(x) \equiv f_j(x)$). However, $f_i(x), i = 1, \dots, n$ here may be very close to each other or even identical infinitely often. This is an essential difficulty arising in the closed-loop identification, compared with [1] where $f_i(x) = x^{b_i}, b_1 > \cdots > b_n > 0$ are quite different when x becomes large. We point out that a trivial extension of the existing methods has trouble coping with the current situation. To overcome this barrier, we establish an inequality on the minimal eigenvalue of the data matrix and the conditional variances of the outputs in Proposition 1. It sets up a bridge between stabilizability and closed-loop identifiability. Based on it, we prove that system (3) is stabilizable if the nonlinear growth rates of $f_1(x), \dots, f_n(x)$ are dominated by some power functions $x^{b_1}, \dots, x^{b_n}$ respectively, where $b_1, \dots, b_n$ satisfy the polynomial rule referred.

Evidently, more unknown parameters means greater uncertainty in system (3). Observe that if the system has a huge

Manuscript received 21 January 2022; accepted 7 June 2022. Date of publication 29 July 2022; date of current version 29 May 2023. This work was supported by the National Natural Science Foundation of China under Grant 11925109 and Grant U21B6001. Recommended by Associate Editor Z. Kan. (Corresponding author: Chanying Li.)

Zhaobo Liu is with the College of Computer Science and Software Engineering, Shenzhen University, Shenzhen 518060, China (e-mail: liuzhaobo15@mails.ucas.ac.cn).

Chanying Li is with the Key Laboratory of Systems and Control, Academy of Mathematics and Systems Science, Chinese Academy of Sciences, Beijing 100190, China, and also with the School of Mathematical Sciences, University of Chinese Academy of Sciences, Beijing 100049, China (e-mail: clyli@amss.ac.cn).

Color versions of one or more figures in this article are available at <https://doi.org/10.1109/TAC.2022.3194962>.

Digital Object Identifier 10.1109/TAC.2022.3194962

Least Squares Model Averaging for Distributed Data

Haili Zhang[†]

*Institute of Applied Mathematics
Shenzhen Polytechnic University
Shenzhen, 518055, China*

ZHANGHL@SZPT.EDU.CN

Zhaobo Liu[†]

*Institute for Advanced Study
Shenzhen University
Shenzhen, 518060, China*

LIUZHAOBO@SZU.EDU.CN

Guohua Zou^{*}

*School of Mathematical Sciences
Capital Normal University
Beijing, 100048, China*

GHZOU@AMSS.AC.CN

Editor: Sayan Mukherjee

Abstract

Divide and conquer algorithm is a common strategy applied in big data. Model averaging has the natural divide-and-conquer feature, but its theory has not been developed in big data scenarios. The goal of this paper is to fill this gap. We propose two divide-and-conquer-type model averaging estimators for linear models with distributed data. Under some regularity conditions, we show that the weights from Mallows model averaging criterion converge in L_2 to the theoretically optimal weights minimizing the risk of the model averaging estimator. We also give the bounds of the in-sample and out-of-sample mean squared errors and prove the asymptotic optimality for the proposed model averaging estimators. Our conclusions hold even when the dimensions and the number of candidate models are divergent. Simulation results and a real airline data analysis illustrate that the proposed model averaging methods perform better than the commonly used model selection and model averaging methods in distributed data cases. Our approaches contribute to model averaging theory in distributed data and parallel computations, and can be applied in big data analysis to save time and reduce the computational burden.

Keywords: consistency, distributed data, divide and conquer algorithm, Mallows' criterion, model averaging, optimality.

1. Introduction

Modern science and technology make data collection easier and easier, and thus more and more big data have been obtained and stored. Usually, such data are with complicated, structured, varied, and various characteristics in economy, finance, biology, medicine, industry, agriculture, transportation, and other fields. See, for example, Misra et al. (2019), who provided plenty of real data examples that reflect the overall outlook of big data era.

. [†] Haili Zhang and Zhaobo Liu contribute equally to this work.

. ^{*} Guohua Zou is corresponding author.

Inverse eigenvalue problem for mass–spring–inerters systems[☆]Zhaobo Liu^a, Qida Xie^{b,c}, Chanying Li^{b,c,*}^a College of Computer Science and Software Engineering of Shenzhen University, Shenzhen 518061, PR China^b Key Laboratory of Systems and Control, Academy of Mathematics and Systems Science, Chinese Academy of Sciences, 100190, PR China^c School of Mathematical Sciences, University of Chinese Academy of Sciences, Beijing 100049, PR China

ARTICLE INFO

Communicated by D. Bernal

Keywords:

Inverse eigenvalue problem
Natural frequency assignment
Inerter
Mass–spring–inerters system
Multiple eigenvalues

ABSTRACT

This paper has solved the inverse eigenvalue problem for a “fixed–free” mass–chain system with inerters. It is well known that for a spring–mass system wherein the adjacent masses are linked through a spring, the natural frequency assignment can be achieved by choosing appropriate masses and spring stiffnesses if and only if the given positive eigenvalues are distinct. However, when we involve inerters, multiple eigenvalues in the assignment are allowed. In fact, for a set of arbitrarily given positive real numbers, we derive a necessary and sufficient condition on the multiplicities of these numbers, which are assigned as the natural frequencies of the concerned mass–spring–inerters system.

1. Introduction

Natural frequency, an inherent attribute of a mechanical vibration system, has attracted wide attention for its importance. In particular, purposefully allocating the natural frequencies to some pre-specified values provides an effective way to induce or evade resonance (see [1,2]). This naturally raises an inverse eigenvalue problem (IEP), that is, to construct a vibration system whose natural frequencies, or mathematically known as eigenvalues, are given beforehand [3–6].

A well-known result on this problem is due to [7] and [8], which is addressed for a mass–spring system. Observe that in such a basic system, the adjacent masses are linked merely by a spring. Therefore, the IEP turns out to be the construction of a Jacobi matrix with its eigenvalues being assigned to a set of specified positive numbers. Employing the tools for Jacobi matrices, [7] and [8] assert that the IEP is solvable if and only if the given positive eigenvalues are distinct. However, multiple eigenvalues often occur in vibration systems, especially in optimized structures [9–15]. As a matter of fact, the IEP admits multiple eigenvalues, if we introduce a mechanical element called the inerter. This new mechanical device can simulate masses by changing its inertance. It was theoretically first studied by [16], completing the force–current analogy between electrical and mechanical networks (see Section II-B of [16]). Through physical realization, inerters have been applied to many engineering fields such as vibration isolators, landing gears, train suspensions, vehicle suspensions, building vibration control, and so on [17–22]. In a mass–spring–inerters system, neighboring masses are linked by a parallel combination of a spring and an inerter. As a starting point, we restrict our interest in this paper to a “fixed–free” system. The term “fixed–free” means one end of the mass–chain system is attached to the ground while the other end is hanging free, as shown in Fig. 1.

The free vibration equation of such a mass–spring–inerters system is described by

$$(\mathbf{M} + \mathbf{B})\ddot{\mathbf{x}} + \mathbf{K}\mathbf{x} = \mathbf{0},$$

[☆] The work was supported by the National Natural Science Foundation of China under Grants 11925109 and 11688101.

* Corresponding author at: Key Laboratory of Systems and Control, Academy of Mathematics and Systems Science, Chinese Academy of Sciences, 100190, PR China.

E-mail address: cyl@amss.ac.cn (C. Li).

A Note on the Convergence of Distributed RLS

Zhaobo Liu  and Chanying Li , Member, IEEE

Abstract—This note focuses on the convergence problem of a distributed recursive least squares (RLS) estimator, which converges fast in many circumstances. We find that the convergence behavior of this distributed RLS is quite different for the scalar-parameter and high-dimensional-parameter cases.

Index Terms—Distributed estimation, mean square convergence, noises, recursive least squares, strong consistency.

I. INTRODUCTION

An adaptive network is built up from a set of nodes, which could communicate with their neighbors through interlinks. Each node observes partial information related to an unknown parameter of common interest and performs local estimation separately [8], [10], [17]. Such network enjoys a certain advantages in robustness and privacy. Specifically, consider a network consisting of n nodes. The corresponding stochastic linear regression models are described by

$$y_{k,i} = \theta^\top \phi_{k,i} + \varepsilon_{k,i}, \quad k \geq 0, \quad i = 1, \dots, n \quad (1)$$

where $\varepsilon_{k,i}$ is a scalar noise sequence and $\theta \in \mathbb{R}^m$ is an unknown non-random parameter vector. At time k , each node i observes a noisy signal $y_{k,i}$ and a data signal $\phi_{k,i}$ taking values in $\mathbb{R}$ and $\mathbb{R}^m$, respectively. Let the network topology be depicted by a directed weighted graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{A})$, where $\mathcal{V} = \{1, 2, \dots, n\}$ is the set of the nodes and $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$ is the set of the edges that any $(i, j) \in \mathcal{E}$ means $\mathcal{G}$ contains a directed path from j to i . The structure of the graph $\mathcal{G}$ is described by the weighted adjacency matrix $\mathcal{A} = \{a_{ij}\}_{n \times n}$, where $a_{ij} > 0$ for $(i, j) \in \mathcal{E}$ and $a_{ij} = 0$ otherwise. Moreover, $\sum_{j=1}^n a_{ij} = 1$ for each $i \in [1, n]$.

At each step k , analogously to the distributed optimization problem [16], we hope to optimize $\sum_{i=1}^n f_{k,i}(x)$, where

$$f_{k,i}(x) = \sum_{j=1}^k (y_{j,i} - \phi_{j,i}^\top x)^2, \quad i = 1, \dots, n$$

are the local loss functions. For the fixed k , [16] achieves it iteratively by minimizing $f_{k,i}$ at each node and then weighting the minimizers from neighbor nodes. However, our problem is a dynamic optimization process with the optimization function varying at each time. This process consumes a lot of computing resources and hence a heuristic algorithm is preferable. Since the recursive least squares (RLS) is one

of the most frequently used algorithms in practice due to its good performance [1], [7], a natural estimation strategy for the network is that the nodes update their estimates according to the RLS rule with the local data and then combine the estimates of their neighbors to produce the final estimates. Mathematically, given initial estimate $\theta_{0,i} \in \mathbb{R}^m$ at node i , the strategy is defined as follows for time k .

1) Adaption:

$$\begin{cases} \beta_{k+1,i} = \theta_{k,i} + L_{k,i}(y_{k,i} - \theta_{k,i}^\top \phi_{k,i}) \\ L_{k,i} = P_{k+1,i} \phi_{k,i} = \frac{P_{k,i} \phi_{k,i}}{1 + \phi_{k,i}^\top P_{k,i} \phi_{k,i}} \\ P_{k+1,i}^{-1} = I_m + \sum_{j=0}^k \phi_{j,i} \phi_{j,i}^\top \end{cases} \quad (2)$$

2) Combination:

$$\theta_{k+1,i} = \sum_{j=1}^n a_{ij} \beta_{k+1,j}. \quad (3)$$

Denote $\tilde{\theta}_{k,i} \triangleq \theta_{k,i} - \theta$ and $\tilde{\Theta}_k \triangleq \text{col}\{\tilde{\theta}_{k,1}, \dots, \tilde{\theta}_{k,n}\}$. Then

$$\tilde{\Theta}_{k+1} = (\mathcal{A} \otimes I_m)(I_{mn} - F_k) \tilde{\Theta}_k + (\mathcal{A} \otimes I_m) L_k V_k,$$

where $\otimes$ is the Kronecker product, $F_k \triangleq L_k \Phi_k^\top$, $L_k \triangleq \text{diag}\{L_{k,1}, \dots, L_{k,n}\}$, $\Phi_k \triangleq \text{diag}\{\phi_{k,1}, \dots, \phi_{k,n}\}$ and $V_k \triangleq \text{col}\{\varepsilon_{k,1}, \dots, \varepsilon_{k,n}\}$. Compared with other network strategies, this estimator converges fast in many circumstances.

Now, given the connectivity of the network, one might guess that if the data at one single node is sufficient to guarantee the strong consistency (mean-square convergence) of the standard RLS, then the distributed RLS (2)–(3) converges to the true parameter accordingly. This guess is confirmed for the scalar-parameter case, where we establish a necessary and sufficient condition on the regression data to guarantee the convergence of the distributed RLS to the true parameter, in the sense of the strong consistency and mean-square convergence. But we cannot take the guess for granted when the unknown parameter is of high dimension. Interestingly, for some “defective” data, the cooperation of the nodes in the connected network causes a diverging error, even if the standard RLS at each node can identify the true parameter by using the local data. According to the “defective” data construction provided later, we find that the distributed RLS (2)–(3) is not suitable for the regression data which vary sharply in the high-dimensional-parameter case. As to the other types of distributed strategies using the RLS, the readers are referred to [3], [9], [18].

II. MAIN RESULTS

We first provide some basic definitions as follows.

Definition 1: (see [13]) A $n \times n$ nonnegative matrix T is irreducible if for every pair i, j of its index set, there exists a positive integer $p = p(i, j)$ such that $T^p[i, j] > 0$, where $T[i, j]$ is the (i, j) th entry of T . An irreducible matrix is said to be aperiodic, if the period of any one (and so of each one) of its indices is equal to 1. T is called stochastic (or more precisely row stochastic) if $\sum_{j=1}^n T[i, j] = 1, \forall i \in [1, n]$. It is called doubly stochastic if also $\sum_{i=1}^n T[i, j] = 1, \forall j \in [1, n]$.

Manuscript received 13 September 2021; accepted 24 November 2021. Date of publication 29 July 2022; date of current version 5 December 2022. This work was supported in part by the National Natural Science Foundation of China under Grant 11925109 and Grant U21B6001 and in part by the Strategic Priority Research Program of Chinese Academy of Sciences under Grant XDA27000000. Recommended by Associate Editor Guoxiang Gu. (Corresponding author: Chanying Li.)

Zhaobo Liu and Chanying Li are with the Key Laboratory of Systems and Control, Academy of Mathematics and Systems Science, Chinese Academy of Sciences, Beijing 100190, China, and also with the School of Mathematical Sciences, University of Chinese Academy of Sciences, Beijing 100049, China (e-mail: liuzhaobo15@mails.ucas.ac.cn; clyli@amss.ac.cn).

Digital Object Identifier 10.1109/TAC.2022.3194882

IS IT POSSIBLE TO STABILIZE DISCRETE-TIME PARAMETERIZED UNCERTAIN SYSTEMS GROWING EXPONENTIALLY FAST?*

ZHAOBO LIU[†] AND CHANYING LI[‡]

Abstract. This paper derives a somewhat surprising but interesting result on the stabilizability of discrete-time parameterized uncertain systems. Contrary to intuition, it shows that the nonlinear growth rate of a discrete-time stabilizable system with linear parameterization is not necessarily small all the time. More specifically, to achieve stabilizability, the system function $f(x) = O(|x|^b)$ with $b < 4$ is only required for a very tiny fraction of x in $\mathbb{R}$, even if it grows exponentially fast for the other x . This x set is called a regular set, whose density determines the stabilizability of a nonlinear discrete-time control system. The densities of regular sets have also been computed for both stabilizable and unstabilizable systems with scalar parameters. As indicated herein, the density could be extremely sparse, while the corresponding system is stabilizable.

Key words. stochastic adaptive control, feedback limitations, stabilizability, nonlinear systems, discrete-time, least squares

AMS subject classifications. 93D15, 93D20, 93E24

DOI. 10.1137/17M1111103

1. Introduction. Linear systems [1], [2], [5], [7] and nonlinear systems with nonlinearities having linear growth rates [24], [26] are studied extensively in adaptive control theory. It is natural that we concentrate here on systems with output nonlinearities growing faster than linearities. Such investigations in the literature are mostly focused on control systems in continuous time [22], [10], [8]. Now comes the noteworthy part. The similarities of adaptive control between continuous- and discrete-time systems no longer exist. A large class of continuous-time nonlinear systems can be globally stabilized by applying nonlinear damping or back-stepping techniques, regardless of how fast its growth rate is [9], [11]. But its discrete-time counterpart is obviously lack of such good property. It was found early in [4] that fundamental difficulties arise for adaptive control of discrete-time nonlinear parameterized systems. Guo [4] proved that any feedback control law may fail to stabilize a discrete-time parameterized system if its nonlinearity is too high. Such a problem also troubles the control of discrete-time nonparametric nonlinear systems [27], [12], [30], semiparametric uncertain systems [6], [23], linear stochastic systems with unknown time-varying parameter processes [28], and continuous-time nonlinear systems with sampled-date observations for prescribed sampling rates [29].

*Received by the editors January 10, 2017; accepted for publication (in revised form) January 31, 2019; published electronically June 13, 2019.

<http://www.siam.org/journals/sicon/57-3/M111110.html>

Funding: This work was supported in part by the National Natural Science Foundation of China under grants 61422308 and 11688101.

[†]Key Laboratory of Systems and Control, Academy of Mathematics and Systems Science, Chinese Academy of Sciences, Beijing 100190, People's Republic of China, and School of Mathematical Sciences, University of Chinese Academy of Sciences, Beijing 100049, People's Republic of China (liuzhaobo15@mails.ucas.ac.cn).

[‡]Corresponding author. Key Laboratory of Systems and Control, Academy of Mathematics and Systems Science, Chinese Academy of Sciences, Beijing 100190, People's Republic of China, and School of Mathematical Sciences, University of Chinese Academy of Sciences, Beijing 100049, People's Republic of China (cyli@amss.ac.cn).

单参数离散时间非线性系统的可镇定性定理

刘兆波, 李婵颖[†]

(中国科学院数学与系统科学研究院 系统科学研究所 系统控制重点实验室, 北京 100080;

中国科学院大学 数学科学院, 北京 100049)

摘要: 本文对基本的离散时间非线性单参数随机系统建立了可镇定性定理. 该定理推进了文献[1]的结果, 进一步完善了关于离散时间自适应控制的反馈能力刻画. 离散时间单参数系统可镇定的一个重要非线性临界常数是4, 用以刻画关于幂函数类系统的反馈能力. 而作为本文定理的应用, 本文对一类典型的单参数离散时间非线性随机系统发现了新的可镇定临界常数2.

关键词: 反馈极限; 自适应控制; 最小二乘法; 可镇定性; 非线性系统; 离散时间

引用格式: 刘兆波, 李婵颖. 单参数离散时间非线性系统的可镇定性定理. 控制理论与应用, 2019, 36(11): 1929 – 1935

DOI: 10.7641/CTA.2019.90640

Stabilizability theorem of discrete-time nonlinear systems with scalar parameters

LIU Zhao-bo, LI Chan-ying[†]

(The Key Laboratory of Systems and Control, Academy of Mathematics and Systems Science,

Chinese Academy of Sciences, Beijing 100190, China;

School of Mathematical Sciences, University of Chinese Academy of Sciences, Beijing 100049, China)

Abstract: This paper advances [1] by deducing a stabilizability theorem for discrete-time nonlinear systems with scalar parameters, which takes a step forward to the complete characterization of feedback limitations in discrete-time adaptive nonlinear control. It is well-known that exponent 4 is an important critical number to characterize the feedback capability for the basic discrete-time scalar-parameter systems, which are governed by power functions. As an application of our theorem, a new critical number 2 is derived for a typical class of discrete-time nonlinear stochastic systems with scalar parameters.

Key words: feedback limitations; adaptive control; least squares; stabilizability; nonlinear systems; discrete time

Citation: LIU Zhaobo, LI Chanying. Stabilizability theorem of discrete-time nonlinear systems with scalar parameters. *Control Theory & Applications*, 2019, 36(11): 1929 – 1935

1 Introduction

Most works on nonlinear adaptive control in the literature are focused on continuous-time systems^[2–4]. But adaptive control between continuous- and discrete-time systems are rather different. As a matter of fact, a large class of continuous-time nonlinear systems can be globally stabilized by applying nonlinear damping or back-stepping techniques, no matter how fast their growth rates are^[5–6]. However, the situation in the discrete-time case is different.

A heuristic result derived by [7] is that feedback limitations exist for discrete-time adaptive nonlinear control. [7] studied a basic discrete-time nonlinear ran-

dom system with a scalar parameter:

$$y_{t+1} = \theta y_t^b + u_t + w_{t+1},$$

and demonstrated that $b = 4$ is the critical exponent for the stabilizability. Soon afterwards [8] established an “impossibility theorem” for the multi-parameter system

$$y_{t+1} = \theta_1 y_t^{b_1} + \theta_2 y_t^{b_2} + \cdots + \theta_n y_t^{b_n} + u_t + w_{t+1}. \quad (1)$$

A polynomial rule on $b_1, \cdots, b_n$ was introduced in the theorem to describe the nonlinear growth rates that fail all feedback control laws in stabilizing system (1). Later, [9] proved that the polynomial rule in fact serves as the necessary and sufficient condition of the stabilizability of system (1). Besides, some initial research

Received 2 August 2019; accepted 15 November 2019.

[†]Corresponding author. E-mail: cyli@amss.ac.cn; Tel.: +86 10-82541631.

Recommended by Associate Editor: LIU Shu-jun.

Supported by the National Key R&D Program of China (2018YFA0703800) and the National Natural Science Foundation of China (11688101).

On the Convergence of Least Squares Estimator for Nonlinear Autoregressive Models*

Zhaobo Liu¹, Chanying Li^{2,3}

1. College of Computer Science and Software Engineering of Shenzhen University, Shenzhen 518060, P. R. China
E-mail: liuzhaobo@szu.edu.cn

2. Key Laboratory of Systems and Control, Academy of Mathematics and Systems Science, Chinese Academy of Sciences, Beijing 100190, P. R. China
E-mail: cyli@amss.ac.cn

3. School of Mathematical Sciences, University of Chinese Academy of Sciences, Beijing 100049, P. R. China

Abstract: This paper is concerned with the least squares estimator for a basic class of nonlinear autoregressive models, whose outputs are not necessarily to be ergodic. Several asymptotic properties of the least squares estimator have been established under mild conditions. These properties suggest the strong consistency of the least squares estimates in nonlinear autoregressive models which are not divergent.

Key Words: Nonlinear Autoregressive Models, Strong Consistency, Least Squares, Harris Recurrent

1 Introduction

System identification is a mature field that has gained a notable attraction in the research community ([1],[4],[12],[13],[15]), however our knowledge of parameter estimation for nonlinear autoregressive models is still developing (e.g. [2],[9],[16]). Consider the following general nonlinear autoregressive (AR) model,

$$y_{t+1} = \theta^\tau \phi(y_t, \dots, y_{t-n+1}) + w_{t+1}, \quad t \geq 0, \quad (1)$$

where θ is the $m \times 1$ unknown parameter vector, y_t, w_t are the scalar observations and random noise signals, and $\phi: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is a known Lebesgue measurable vector function, respectively. In the literature, asymptotic theory has been established for estimating nonlinear AR models based on an important assumption that the underlying systems satisfy certain ergodic or stationary conditions. Since the least squares (LS) estimator is one of the most basic and effective methods in parameter estimation, the goal of this article is to estimate parameter θ in model (1) by using LS estimator, no matter the outputs are ergodic or not.

Recently, [11] established some asymptotic properties of the LS estimator for model (1) by restricting ϕ as

$$\phi(z_1, \dots, z_n) = \text{col}\{\phi^{(1)}(z_1), \dots, \phi^{(n)}(z_n)\}, \quad (2)$$

where $\phi^{(i)} = (f_{i1}, \dots, f_{im_i})^\tau: \mathbb{R} \rightarrow \mathbb{R}^{m_i}, i = 1, \dots, n$ are some known Lebesgue measurable vector functions and $m_i \geq 1$ are n integers satisfying $\sum_{i=1}^n m_i = m$. [11] shows that the LS estimates converge to the true parameter almost surely on the set where vectors $(y_t, \dots, y_{t-n+1})^\tau$ do not diverge to infinity. This means the LS estimator is very likely to be strong consistency when applied to model (1)–(2) in practice owing to that most real systems are not divergent. It is worth nothing that [7] has completely solved the conver-

gence problem of the LS estimator for the linear AR model

$$y_{t+1} = \sum_{i=1}^n \theta_i y_{t-i+1} + w_{t+1},$$

which is a special case of model (1)–(2).

As to nonlinear model (1) without restriction (2), we naturally wonder if the LS estimator still has a similar asymptotic behavior. Indeed, there are some situations that (2) does not hold, such as exponential autoregressive model (EAR) or threshold autoregressive model (TAR) (see [2],[3]). Now, define

$$P_{t+1}^{-1} = I_m + \sum_{i=0}^t \phi(y_i, \dots, y_{i-n+1}) \phi(y_i, \dots, y_{i-n+1})^\tau,$$

which is referred to as “information matrix”. The key of studying the strong consistency of the LS for model (1) is to estimate the growth rate of the minimal eigenvalue of P_{t+1}^{-1} ([5],[6],[8],[14]). In Section 2, under some mild assumptions, the minimal eigenvalue of P_{t+1}^{-1} are estimated in both the Gaussian case and non-Gaussian cases, respectively. The proofs of the main results are included in Section 3.

2 Main Results

We shall discuss the parameter estimation of model (1) by treating θ as a random variable and a constant vector, respectively. Firstly, consider model (1) under the Gaussian case. Assume

- A1** The noise $\{w_t\}$ is an i.i.d random sequence with $w_1 \sim N(0, 1)$ and parameter $\theta \sim N(\theta_0, I_m)$ is independent of $\{w_t\}$.
- A2** There is a bounded open set $E \subset \mathbb{R}^n$ and a number $\delta^* > 0$ such that
 - (i) $f_i \in C(\mathbb{R}^n), 1 \leq i \leq m;$
 - (ii) for every unit vector $x \in \mathbb{R}^n$,

$$J(\{y \in \bar{E} : |\phi^\tau(y)x| = \delta^*\}) = 0, \quad (3)$$

and

$$\inf_{\|x\|=1} \ell(\{y \in E : |\phi^\tau(y)x| > \delta^*\}) > 0, \quad (4)$$

*This work is supported by the National Natural Science Foundation (NNSF) of China under Grant 11925109 and the National Key R&D Program of China (2018YFA0703800).

*Corresponding author: Chanying Li.